

## Table of Contents

|                                     |    |
|-------------------------------------|----|
| Introduction .....                  | 2  |
| Addressing table.....               | 3  |
| Network Topology (Logical).....     | 9  |
| Physical design .....               | 11 |
| Layer 2 Design and features.....    | 15 |
| Layer 3 Design and features.....    | 16 |
| WAN/VPN Design .....                | 17 |
| Presentation Layer .....            | 17 |
| Security Services Layer.....        | 18 |
| Architecture Diagrams – HQ .....    | 20 |
| Architecture Diagrams – BR .....    | 21 |
| Deployment Diagram – HQ.....        | 22 |
| Deployment Diagram – BR.....        | 23 |
| Deployment Diagram – WAN/IPSec..... | 24 |

## Introduction

This section outlines the detailed network design for the proposed solution for BAK Group focusing on the implementation of a secure, reliable, high availability and centralized services for a network enterprise infrastructure.

The architecture for the proposed network integrates both Local Area Network and Wide Area Network using hierarchical design model. The used core technologies in the design are IPsec VPN and OSPF for the internal dynamic routing protocol in addition to the VLAN segmentation while enforcing the security using firewalls and other security mechanisms.

Additionally, the centralized services are deployed on the headquarter site and it includes Active Directory and Network Policy Server for authentication and access control management, DNS and DHCP as a supporting service for end users, Syslog for centralized collecting and storing device logs, monitoring for centralized monitoring activity and availability of the enter network infrastructure.

This document provides a detailed description of the network design including all its feature which will be distributed into logical topology, addressing scheme, Layer 2 and Layer 3 design decisions, security architecture. This document intended for network engineers and system administrators for future implementing, maintaining and troubleshooting.

## Addressing table

| Summary IPv4 Address |                 |
|----------------------|-----------------|
| Area                 | Network Address |
| HQ                   | 172.17.0.0/16   |
| Branch               | 172.18.0.0/16   |
| ISP                  | 50.0.0.0/26     |

Table 1 Appendix 2 - Summary IPv4 Table

| VLAN Table for HQ |                       |                 |                         |
|-------------------|-----------------------|-----------------|-------------------------|
| VLAN              | Name                  | IP Address      | Usage                   |
| 10                | Sales                 | 172.17.10.0/24  | For sales VLAN          |
| 20                | IT                    | 172.17.20.0/24  | For IT VLAN             |
| 30                | HR                    | 172.17.30.0/24  | For HR VLAN             |
| 40                | Purchase              | 172.17.40.0/24  | For purchase VLAN       |
| 50                | Accounting            | 172.17.50.0/24  | For accounting VLAN     |
| 99                | Switch Native         | N/A             | Native VLAN on switches |
| 100               | Servers Devices       | 172.17.100.0/24 | For Server Only         |
| 254               | Management Interfaces | 172.17.254.0/24 | Manage the devices      |
| 255               | ASA-Failover          | 172.17.255.0/24 | ASA Failover Links      |
| 256               | WAN Side              | N/A             | WAN side communication  |
| 999               | Black Hole            | N/A             | Unused port             |

Table 2 Appendix 2 - VLAN Table for HQ

| VLAN Table for BR |                       |                 |                         |
|-------------------|-----------------------|-----------------|-------------------------|
| VLAN              | Name                  | IP Address      | Usage                   |
| 10                | Sales                 | 172.18.10.0/24  | For sales VLAN          |
| 20                | IT                    | 172.18.20.0/24  | For IT VLAN             |
| 30                | HR                    | 172.18.30.0/24  | For HR VLAN             |
| 40                | Purchase              | 172.18.40.0/24  | For purchase VLAN       |
| 50                | Accounting            | 172.18.50.0/24  | For accounting VLAN     |
| 99                | Switch Native         | N/A             | Native VLAN on switches |
| 100               | Servers Devices       | 172.18.100.0/24 | For Server Only         |
| 254               | Management Interfaces | 172.18.254.0/24 | Manage the devices      |
| 255               | ASA-Failover          | 172.18.255.0/24 | ASA Failover Links      |
| 256               | WAN Side              | N/A             | WAN side communication  |
| 999               | Black Hole            | N/A             | Unused port             |

Table 3 Appendix 2 - VLAN Table for New Branch

| HQ IP Address Table |                      |               |                 |                 |
|---------------------|----------------------|---------------|-----------------|-----------------|
| Device              | Interface            | IP Address    | Subnet Mask     | Default Gateway |
| HQ-Router           | G0/1 (ISP-R5)        | 50.0.0.54     | 255.255.255.252 | N/A             |
|                     | G0/2 (ISP-R2)        | 50.0.0.50     | 255.255.255.252 | N/A             |
| HQ-FW1              | G0/0 (Inside-Switch) | 172.17.255.1  | 255.255.255.0   | N/A             |
|                     | G0/3 (WAN-Side)      | 50.0.0.1      | 255.255.255.252 | N/A             |
| HQ-FW2              | G0/0 (Inside-Switch) | 172.17.255.1  | 255.255.255.0   | N/A             |
|                     | G0/3 (WAN-Side)      | 50.0.0.1      | 255.255.255.252 | N/A             |
| HQ-CSW1             | G0/1 (HQ-CSW2)       | 172.17.0.13   | 255.255.255.252 | N/A             |
|                     | VLAN 10              | 172.17.10.1   | 255.255.255.0   | N/A             |
|                     | VLAN 20              | 172.17.20.1   | 255.255.255.0   | N/A             |
|                     | VLAN 30              | 172.17.30.1   | 255.255.255.0   | N/A             |
|                     | VLAN 40              | 172.17.40.1   | 255.255.255.0   | N/A             |
|                     | VLAN 50              | 172.17.50.1   | 255.255.255.0   | N/A             |
|                     | VLAN 100             | 172.17.100.1  | 255.255.255.0   | N/A             |
|                     | VLAN 254             | 172.17.254.1  | 255.255.255.0   | N/A             |
|                     | VLAN 255             | 172.17.255.4  | 255.255.255.0   | N/A             |
| HQ-CSW2             | G0/1 (HQ-CSW2)       | 172.17.0.14   | 255.255.255.252 | N/A             |
|                     | VLAN 10              | 172.17.10.2   | 255.255.255.0   | N/A             |
|                     | VLAN 20              | 172.17.20.2   | 255.255.255.0   | N/A             |
|                     | VLAN 30              | 172.17.30.2   | 255.255.255.0   | N/A             |
|                     | VLAN 40              | 172.17.40.2   | 255.255.255.0   | N/A             |
|                     | VLAN 50              | 172.17.50.2   | 255.255.255.0   | N/A             |
|                     | VLAN 100             | 172.17.100.2  | 255.255.255.0   | N/A             |
|                     | VLAN 254             | 172.17.254.2  | 255.255.255.0   | N/A             |
|                     | VLAN 255             | 172.17.255.5  | 255.255.255.0   | N/A             |
| HQ-DSW1             | VLAN 254             | 172.17.254.13 | 255.255.255.0   | 172.17.254.254  |
| HQ-DSW2             | VLAN 254             | 172.17.254.14 | 255.255.255.0   | 172.17.254.254  |
| HQ-ASW1             | VLAN 254             | 172.17.254.15 | 255.255.255.0   | 172.17.254.254  |
| HQ-ASW2             | VLAN 254             | 172.17.254.16 | 255.255.255.0   | 172.17.254.254  |
| HQ-ASW3             | VLAN 254             | 172.17.254.17 | 255.255.255.0   | 172.17.254.254  |
| HQ-SerFarm1         | VLAN 254             | 172.17.254.18 | 255.255.255.0   | 172.17.254.254  |
| HQ-SerFarm2         | VLAN 254             | 172.17.254.19 | 255.255.255.0   | 172.17.254.254  |
| Services Server     | E0                   | 172.17.100.12 | 255.255.255.0   | 172.17.100.254  |
|                     | E1                   |               |                 |                 |
| DC01                | E0                   | 172.17.100.13 | 255.255.255.0   | 172.17.100.254  |
|                     | E1                   |               |                 |                 |
| DC02                | E0                   | 172.17.100.14 | 255.255.255.0   | 172.17.100.254  |
|                     | E1                   |               |                 |                 |
| PC1                 | E0                   | DHCP          |                 |                 |
| PC2                 | E0                   | DHCP          |                 |                 |
| PC3                 | E0                   | DHCP          |                 |                 |

|     |    |      |
|-----|----|------|
| PC4 | E0 | DHCP |
| PC5 | E0 | DHCP |

Table 4 Appendix 2 - HQ Addressing Table

| New Branch IP Address Table |                      |               |                 |                 |
|-----------------------------|----------------------|---------------|-----------------|-----------------|
| Device                      | Interface            | IP Address    | Subnet Mask     | Default Gateway |
| BR-Router                   | G0/1 (ISP-R1)        | 50.0.0.58     | 255.255.255.252 | N/A             |
|                             | G0/2 (ISP-R2)        | 50.0.0.62     | 255.255.255.252 | N/A             |
| BR-FW1                      | G0/0 (Inside-Switch) | 172.17.255.1  | 255.255.255.0   | N/A             |
|                             | G0/3 (WAN-Side)      | 50.0.0.1      | 255.255.255.252 | N/A             |
| BR-FW2                      | G0/0 (Inside-Switch) | 172.17.255.1  | 255.255.255.0   | N/A             |
|                             | G0/3 (WAN-Side)      | 50.0.0.1      | 255.255.255.252 | N/A             |
| BR-CSW1                     | G0/1 (BR-CSW2)       | 172.18.0.13   | 255.255.255.252 | N/A             |
|                             | VLAN 10              | 172.18.10.1   | 255.255.255.0   | N/A             |
|                             | VLAN 20              | 172.18.20.1   | 255.255.255.0   | N/A             |
|                             | VLAN 30              | 172.18.30.1   | 255.255.255.0   | N/A             |
|                             | VLAN 40              | 172.18.40.1   | 255.255.255.0   | N/A             |
|                             | VLAN 50              | 172.18.50.1   | 255.255.255.0   | N/A             |
|                             | VLAN 100             | 172.18.100.1  | 255.255.255.0   | N/A             |
|                             | VLAN 254             | 172.18.254.1  | 255.255.255.0   | N/A             |
|                             | VLAN 255             | 172.18.255.4  | 255.255.255.0   | N/A             |
| BR-CSW2                     | G0/1 (BR-CSW1)       | 172.18.0.14   | 255.255.255.252 | N/A             |
|                             | VLAN 10              | 172.18.10.2   | 255.255.255.0   | N/A             |
|                             | VLAN 20              | 172.18.20.2   | 255.255.255.0   | N/A             |
|                             | VLAN 30              | 172.18.30.2   | 255.255.255.0   | N/A             |
|                             | VLAN 40              | 172.18.40.2   | 255.255.255.0   | N/A             |
|                             | VLAN 50              | 172.18.50.2   | 255.255.255.0   | N/A             |
|                             | VLAN 100             | 172.18.100.2  | 255.255.255.0   | N/A             |
|                             | VLAN 254             | 172.18.254.2  | 255.255.255.0   | N/A             |
|                             | VLAN 255             | 172.18.255.5  | 255.255.255.0   | N/A             |
| BR-DSW1                     | VLAN 254             | 172.18.254.13 | 255.255.255.0   | 172.18.254.254  |
| BR-DSW2                     | VLAN 254             | 172.18.254.14 | 255.255.255.0   | 172.18.254.254  |
| BR-ASW1                     | VLAN 254             | 172.18.254.15 | 255.255.255.0   | 172.18.254.254  |
| BR-ASW2                     | VLAN 254             | 172.18.254.16 | 255.255.255.0   | 172.18.254.254  |
| BR-ASW3                     | VLAN 254             | 172.18.254.17 | 255.255.255.0   | 172.18.254.254  |
| BR-SerFarm1                 | VLAN 254             | 172.18.254.18 | 255.255.255.0   | 172.17.254.254  |
| BR-SerFarm2                 | VLAN 254             | 172.18.254.19 | 255.255.255.0   | 172.17.254.254  |
| DC03                        | E0                   | 172.18.100.13 | 255.255.255.0   | 172.18.100.254  |
|                             | E1                   |               |                 |                 |
| PC1                         | E0                   | DHCP          |                 |                 |
| PC2                         | E0                   | DHCP          |                 |                 |

|     |    |      |
|-----|----|------|
| PC3 | E0 | DHCP |
| PC4 | E0 | DHCP |
| PC5 | E0 | DHCP |

Table 5 Appendix 2 - New Branch Addressing Table

| ISP IP Address Table |                  |            |                 |                 |
|----------------------|------------------|------------|-----------------|-----------------|
| Device               | Interface        | IP Address | Subnet Mask     | Default Gateway |
| ISP-R1               | E0/0 (ISP-R4)    | 50.0.0.17  | 255.255.255.252 | N/A             |
|                      | E0/1 (ISP-R2)    | 50.0.0.21  | 255.255.255.252 | N/A             |
|                      | E0/2 (ISP-R3)    | 50.0.0.25  | 255.255.255.252 | N/A             |
|                      | E0/3 (BR-Router) | 50.0.0.57  | 255.255.255.252 | N/A             |
|                      | Lo0              | 1.1.1.1    | 255.255.255.255 | N/A             |
| ISP-R2               | E0/0 (ISP-R5)    | 50.0.0.29  | 255.255.255.252 | N/A             |
|                      | E0/1 (ISP-R1)    | 50.0.0.22  | 255.255.255.252 | N/A             |
|                      | E0/2 (HQ-Router) | 50.0.0.49  | 255.255.255.252 | N/A             |
|                      | E0/3 (BR-Router) | 50.0.0.61  | 255.255.255.252 | N/A             |
|                      | E1/0 (ISP-R3)    | 50.0.0.33  | 255.255.255.252 | N/A             |
|                      | Lo0              | 2.2.2.2    | 255.255.255.255 | N/A             |
| ISP-R3               | E0/0 (ISP-R2)    | 50.0.0.34  | 255.255.255.252 | N/A             |
|                      | E0/1 (ISP-R4)    | 50.0.0.37  | 255.255.255.252 | N/A             |
|                      | E0/2 (ISP-R1)    | 50.0.0.26  | 255.255.255.252 | N/A             |
|                      | E0/3 (ISP-R5)    | 50.0.0.41  | 255.255.255.252 | N/A             |
|                      | Lo0              | 3.3.3.3    | 255.255.255.255 | N/A             |
| ISP-R4               | E0/0 (ISP-R1)    | 50.0.0.18  | 255.255.255.252 | N/A             |
|                      | E0/1 (ISP-R5)    | 50.0.0.45  | 255.255.255.252 | N/A             |
|                      | E0/2 (ISP-R3)    | 50.0.0.38  | 255.255.255.252 | N/A             |
|                      | Lo0              | 4.4.4.4    | 255.255.255.255 | N/A             |
| ISP-R5               | E0/0 (ISP-R2)    | 50.0.0.30  | 255.255.255.252 | N/A             |
|                      | E0/1 (ISP-R4)    | 50.0.0.46  | 255.255.255.252 | N/A             |
|                      | E0/2 (HQ-Router) | 50.0.0.53  | 255.255.255.252 | N/A             |
|                      | E0/3 (ISP-R3)    | 50.0.0.42  | 255.255.255.252 | N/A             |
|                      | Lo0              | 5.5.5.5    | 255.255.255.255 | N/A             |

Table 6 Appendix 2 - ISP Addressing Table

| HQ Router IDs |           |
|---------------|-----------|
| Device Name   | Router ID |
| HQ-FW1        | 1.1.1.1   |
| HQ-CSW1       | 1.1.2.1   |
| HQ-CSW2       | 1.1.2.2   |

Table 7 Appendix 2 - HQ OSPF Router-ID Table

| BR Router IDs |           |
|---------------|-----------|
| Device Name   | Router ID |
| BR-FW1        | 2.2.1.1   |
| BR-CSW1       | 2.2.2.1   |
| BR-CSW2       | 2.2.2.2   |

Table 8 Appendix 2 - New Branch OSPF Router-ID Table

| Local AAA Username & Password |             |                |
|-------------------------------|-------------|----------------|
| Device                        | Username    | Password       |
| HQ-FW1                        | HQ-FW1      | hqfw1@ssh      |
| HQ-CSW1                       | HQ-CSW1     | hqcsw1@ssh     |
| HQ-CSW2                       | HQ-CSW2     | hqcsw2@ssh     |
| HQ-DSW1                       | HQ-DSW1     | hqds1@ssh      |
| HQ-DSW2                       | HQ-DSW2     | hqds2@ssh      |
| HQ-ASW1                       | HQ-ASW1     | hqas1@ssh      |
| HQ-ASW2                       | HQ-ASW2     | hqas2@ssh      |
| HQ-ASW3                       | HQ-ASW3     | hqas3@ssh      |
| HQ-SerFarm1                   | HQ-SerFarm1 | hqserfarm1@ssh |
| HQ-SerFarm2                   | HQ-SerFarm2 | hqserfarm2@ssh |
| BR-FW1                        | BR-FW1      | brfw1@ssh      |
| BR-CSW1                       | BR-CSW1     | brcsw1@ssh     |
| BR-CSW2                       | BR-CSW2     | brcsw2@ssh     |
| BR-DSW1                       | BR-DSW1     | brdsw1@ssh     |
| BR-DSW2                       | BR-DSW2     | brdsw2@ssh     |
| BR-ASW1                       | BR-ASW1     | brasw1@ssh     |
| BR-ASW2                       | BR-ASW2     | brasw2@ssh     |
| BR-ASW3                       | BR-ASW3     | brasw3@ssh     |
| BR-SerFarm1                   | BR-SerFarm1 | brserfarm1@ssh |
| BR-SerFarm2                   | BR-SerFarm2 | brserfarm2@ssh |

Table 9 Appendix 2 - Local AAA Username & Password

| AAA Server Username & Passwords |               |
|---------------------------------|---------------|
| Username                        | Password      |
| HAli                            | BakGroup@2026 |
| AKhalid                         | BakGroup@2026 |

Table 10 Appendix 2 - Centralized AAA Username & Password

| Enable Passwords |          |
|------------------|----------|
| Device           | Password |

|             |                 |
|-------------|-----------------|
| HQ-FW1      | p@ssw0rd        |
| HQ-CSW1     | hqcsw1@2026     |
| HQ-CSW2     | hqcsw2@2026     |
| HQ-DSW1     | hqdsww1@2026    |
| HQ-DSW2     | hqdsww2@2026    |
| HQ-ASW1     | hqasw1@2026     |
| HQ-ASW2     | hqasw2@2026     |
| HQ-ASW3     | hqasw3@2026     |
| HQ-SerFarm1 | hqserfarm1@2026 |
| HQ-SerFarm2 | hqserfarm2@2026 |
| BR-FW1      | p@ssw0rd        |
| BR-CSW1     | brcsw1@2026     |
| BR-CSW2     | brcsw2@2026     |
| BR-DSW1     | brdsw1@2026     |
| BR-DSW2     | brdsw2@2026     |
| BR-ASW1     | brasw1@2026     |
| BR-ASW2     | brasw2@2026     |
| BR-ASW3     | brasw3@2026     |
| BR-SerFarm1 | brserfarm1@2026 |
| BR-SerFarm2 | brserfarm2@2026 |

Table 11 Appendix 2 - Local Enable Passwords

| VTP         |              |
|-------------|--------------|
| Domain name | BAKGroup.com |
| Domain pass | BAK@Group    |

Table 12 Appendix 2 - VTP Table

## Network Topology (Logical)

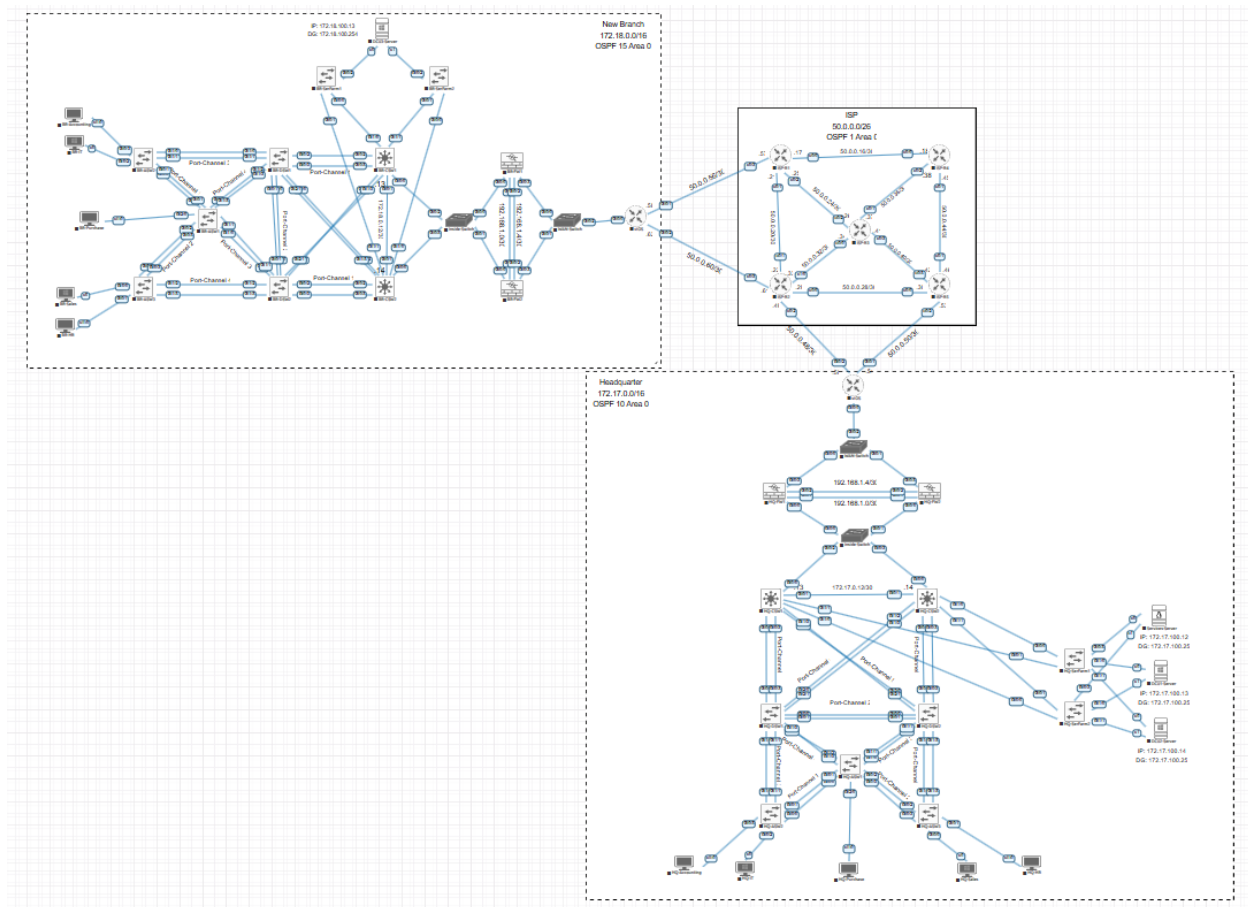
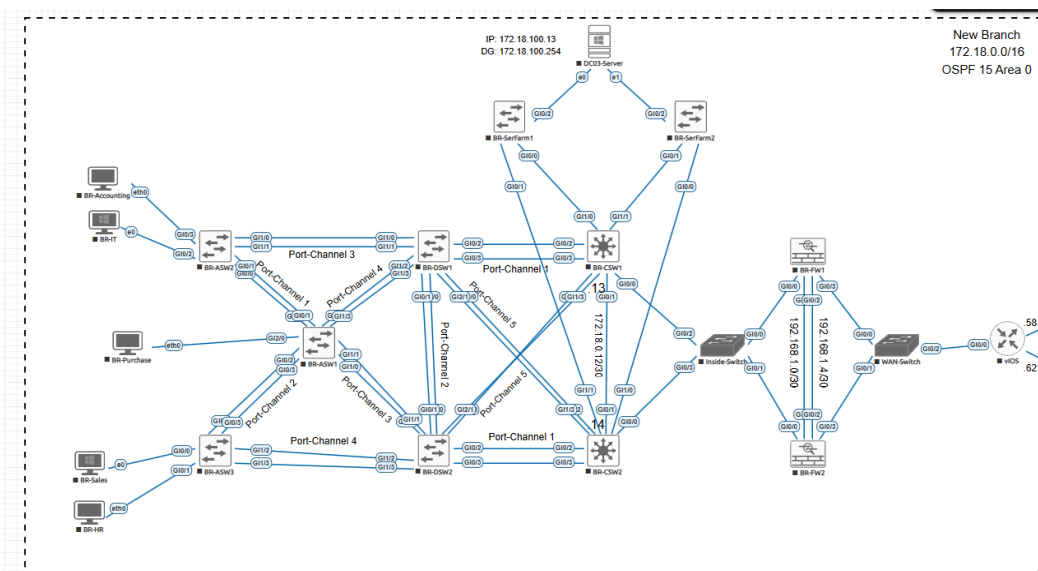
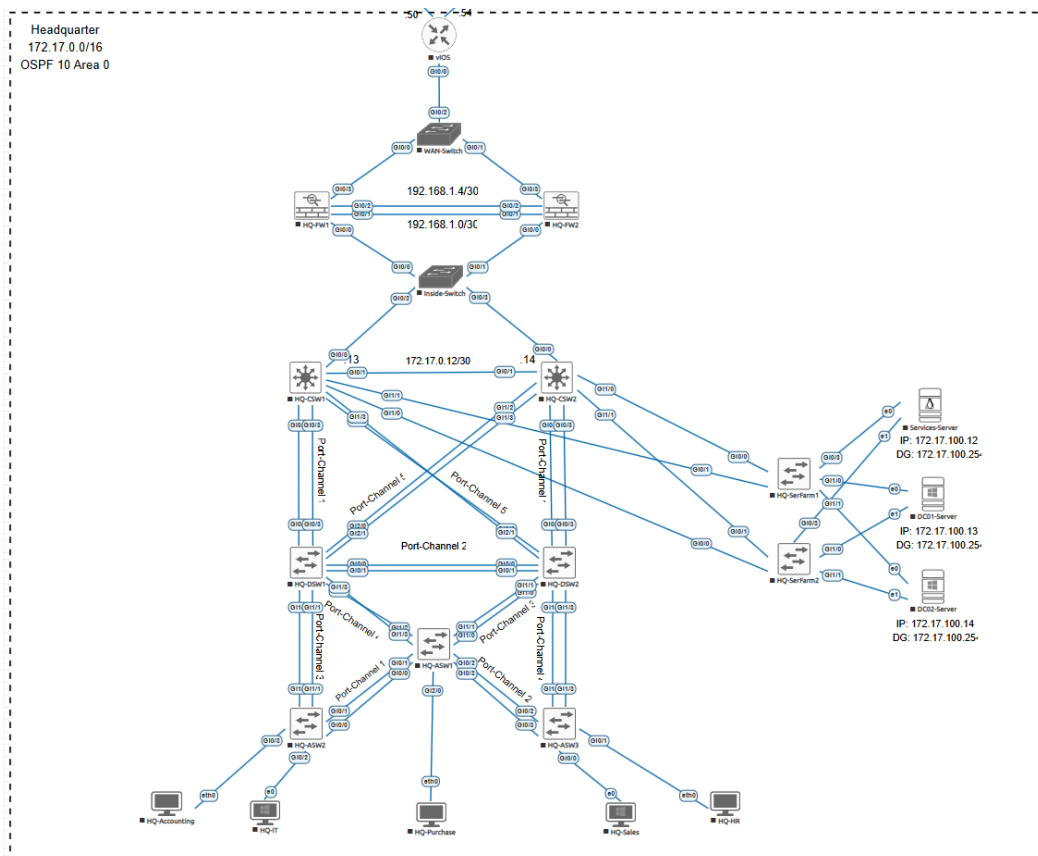


Figure 1 Appendix 2 - Logical Topology



## Physical design

This section will provide an explanation for the physical rack design for each site including the HQ and New branch of BAK Group **for any future physical deployment.**

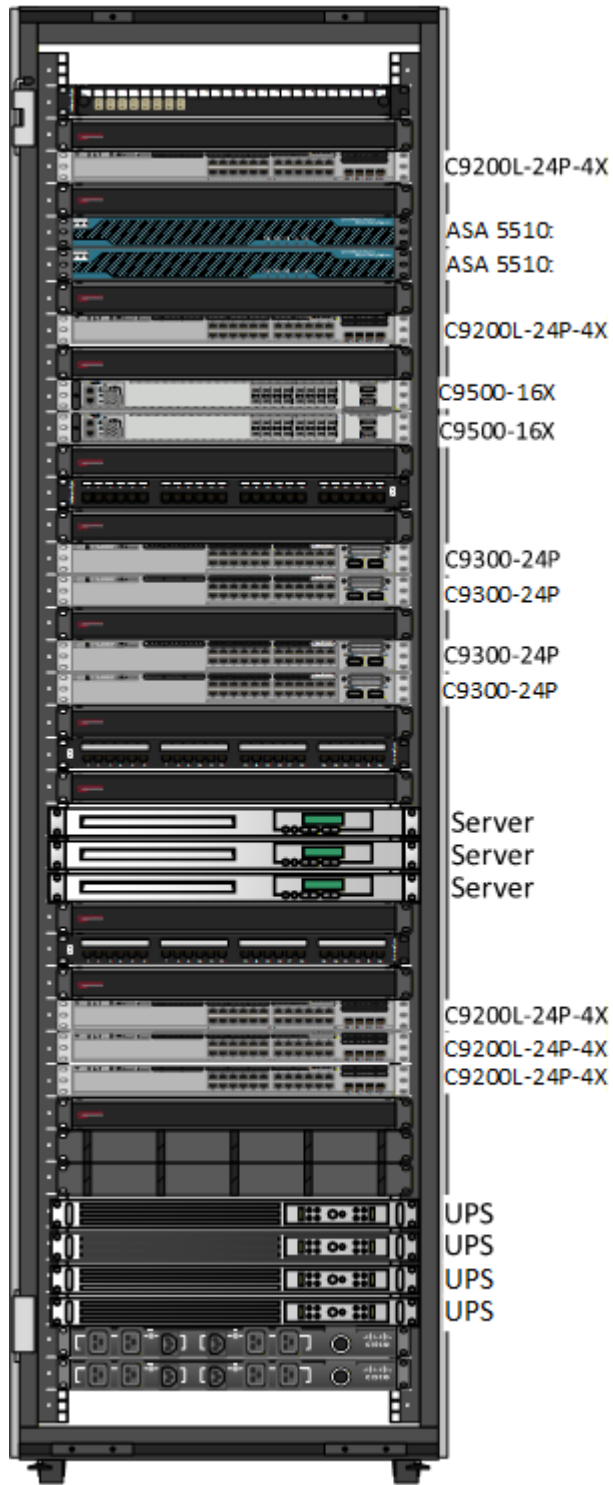


Figure 4 Appendix 2 - Physical Rack Design HQ

The BAK Group Headquarter rack design show the physical placement and organization of the network devices, servers, cable management and the power equipment based on the proposed topology.

**The upper section** of the rack is mainly used for the incoming cables from the ISP alongside with the wan switch and the ASA firewall pair which is mainly used for external connectivity, security and VPN establishment between the HQ and New Branch. Below the firewalls **comes the inside switch and the C9500 core switches** which are the responsible section for internal routing. **The server farm section** uses two of the C9300 switches connected directly to the C9500 core switches allowing for more organized and robust redundancy design and connectivity for the servers.

**The distribution switches** are also using the C9300 series and it placed below the core layer switches to aggregate the connection towards the C9200 access switches. **The C9200 access switches** is mainly to support the end user devices such as Laptops, Desktops and Printer.

**Horizontal cable management** units are installed between each major section to keep cabling structured, organized and to improve troubleshooting in addition to allowing for future expansion. **At the bottom of the rack**, the UPS units, PDUs, grounding, and cable entry provide stable power distribution, backup power, and better rack stability.

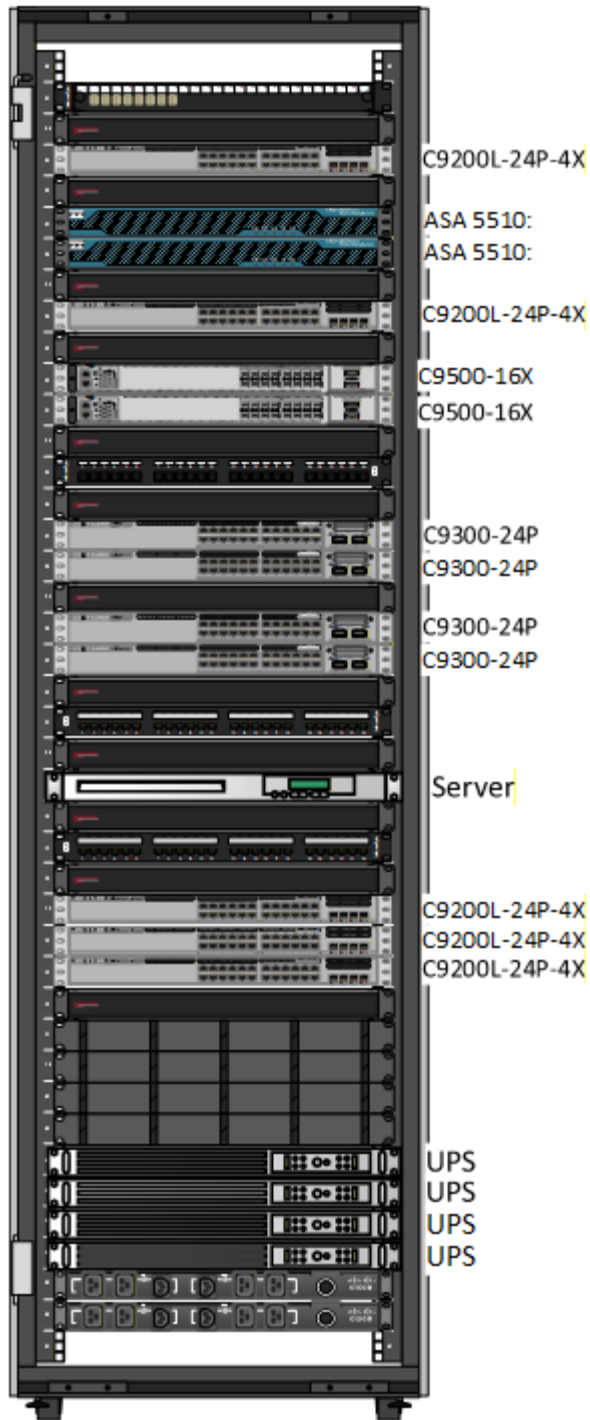


Figure 5 Appendix 2 - Physical Rack Design New Branch

The new branch of BAK Group uses the same structured design that are used in the headquarter but it is scaled to match the smaller requirements of the new branch.

**The upper section** of the rack contains the WAN connection, the wan switch and ASA pair which they provide the secure edge VPN connectivity between the branch and HQ in terms of VPN and between the branch and the outside network in terms in normal network connectivity.

Below the firewall pair, **the inside switch and C9500 core switches** are mounted which are mainly providing the internal routing layer for the new branch. **The new branch server farm** section is reduced to support the lower requirement by using a single local server which is connected through the C9300 server farm switches to the core switches layer for reliable internal service access.

**The C9300 distribution switches** aggregate connections toward the **C9200 access switches**, which provide connectivity for users and endpoint devices through the access patch panel. **Horizontal cable management** units are placed between major rack sections to maintain organized cabling, improve troubleshooting, and support future expansion.

**At the bottom of the rack** the UPS units, PDUs, grounding, and cable entry provide stable power distribution, backup power, and rack stability

## Layer 2 Design and features

The layer 2 design is mainly focuses on providing efficient switching loop prevention and network segmentation inside the network. Layer 2 features used in the system are VLAN, STP and EtherChannel.

**VLAN segmentation** were implemented to logically separates the different departments which will improve both security and performance by reducing the number of broadcast domains. Each of the VLANs were assigned a unique identifier, an entire subnet and a purpose based on the company requirements.

**Spanning Tree Protocol (STP)** was configured to prevent the switching loops and ensure that the network is always stable by maintain a loop-free topology. In addition to the STP protection toolkit which consist of root guard, loop guard and BPDU guard were implemented to enhance and improve the reliability of the network and prevent misconfiguration or unauthorized devices from affecting and changing the organization STP topology. **VLAN Trunking Protocol (VTP)** was also used to simplify the VLAN management in the entire company by allowing the VLAN information to propagate and transfer across al the switches that are under the same domain.

**EtherChannel** was used to combine and aggregate multiple links into a single logical link which will result in increasing the link bandwidth in addition to providing and ensuring redundancy. This ensures that if one link fails, traffic can continue to flow through the remaining links without disruption. LACP was selected over the other available protocol such as PAgP due to its status as an open-source protocol which ensures that it's supported across different vendors devices. on the other hand, PAgP is a Cisco proprietary protocol and its limited to Cisco devices only. LACP allows greater flexibility and scalability in multi-vendor networks. LACP offers more advanced control over link aggregation, making it more suitable for enterprise network designs.

## Layer 3 Design and features

The layer 3 design is mainly focused on routing functionality and inter-network communication across the entire enterprise network. Layer 3 features used in the system are OSPF, GLBP and inter-VLAN routing.

**Open Shortest Path First (OSPF) protocol** was configured as the dynamic routing protocol to enable and allows automatic route exchanged between network devices. OSPF considered to a link state protocol and it allow the network to adopt to any changes within the networks in addition to that OSPF ensures optimal path selection for data transmissions and exchange. OSPF was selected over the other alternatives such as EIGRP or RiP due to its scalability and open standards nature. While RiP protocol is simple to configure it has a significant limitation when it comes to network size and coverage speed making it unsuitable for scalable enterprise network. On the other hand, EIGRP provides efficient routing and fast coverage but it is primarily a Cisco proprietary protocol which limits the compatibility with other protocols and vendors. OSPF is an open standard protocol which offers better flexibility, supports large-scale networks, and ensures compatibility across different platforms that why OSPF is the best choice and more suitable.

**Gateway Load Balancing Protocol (GLBP)** was implemented to provides default gateway redundancy and load balancing at the same time. This is to ensure high availability by allowing multiple devices to share a single virtual IP to load balance the incoming traffic between them while also providing automatic failover in case of any device failure. GLBP was chosen over the alternatives such as HSRP and VRRP due its ability to provide both load balancing and redundancy at the same time natively. While HSRP and VRRP only offers gateway redundancy they do not support load balancing between multiple devices. GLBP allows efficient usage of network resources by distributing traffic across multiple gateways that why it more suitable that the other alternatives.

**Inter-VLAN routing** was implemented to allow communication between different VLANs ensuring data exchange between the authorized VLANs only ensuring controlled and efficient data flow.

## WAN/VPN Design

**The WAN design connects** the BAK headquarters and new branch through a simulated ISP network inside EVE-NG. A Site-to-Site IPsec VPN was configured to ensure secure and encrypted communications between the company branches over the simulated public ISP WAN connection. The VPN tunnel encrypts all the transmitted data between the sites and maintain confidentiality and integrity.

The IPsec tunnels were configured on the firewall devices at both location with a solid defines encryption and authentication parameters. This approach ensures that only authorized devices can establish an VPN tunnel connection.

**IPsec Site-to-Site VPN was selected** over the alternative such as GRE or SSL VPN due to its strong security, authentication and its suitability for permanent inter-branch connectivity while the GRE provides flexibility in transporting different protocol within the tunnel it does not offer any encryption by default making it less secure for sensitive data transfer and more work to configure encryption onto the GRE tunnel. The SSL VPN is more suitable and designed for remote access users rather than full site-to-site connectivity. on the other hand, IPsec provides robust encryption, authentication, and data integrity, ensuring secure and reliable communication between the headquarters and branch networks making it more suitable for this project.

## Presentation Layer

The presentation layer focused on the user interaction with all network services on the infrastructure. End user services were configured to support the organizational operations including email communication and centralized authentication.

The SMTP relay server was configured to handle internal system generated notifications, such as PRTG alerts while user and external email communication will use Microsoft Outlook services. This approach ensures that system notifications and day-to-day communication are separated, improving security, reliability, and scalability compared to a single unified solution. Additionally, Active Directory were deployed to provides centralized authentication to enables the users to access the available network resources based on the defined policies and their authorization level.

Protocols such as SMTP and HTTP/HTTPS were used to allows communication between users and services. This implementation ensures a user-friendly and secure environment for accessing the internal network resources of the company.

## Security Services Layer

Security considered be to a critical aspect of the network design and to achieve maximum security multiple layers of protection were implemented across the infrastructure. Security layers used in the project are Cisco ASA, IPsec VPN, Centralized authentication, monitoring and logging in addition to a multi-layer ACL, VPN-Filters, Network and VLAN segmentation and layer 2 security.

**Cisco ASA** firewall was deployed at the network edge to control the incoming and outgoing traffic between the sites and external networks. The firewall enforces security by filtering the traffic based on the configured rules and policies which prevents unauthorized access to the internal resources of the network. The firewall also acts as the primary gateway to the public WAN while using both NAT and VPN configuration. This ensures that all external communication is inspected and controlled by the ASA acting as the first line of defense for the infrastructure.

**The IPsec VPN** was used to provides an encrypted communication between the headquarters and new branch network. The IPsec VPN ensures data confidentiality, integrity and authentication while the data being transmitting across the public WAN by using encryption algorithms and key exchange mechanisms such as IKEv2. This approach ensures that the company sensitive data are protected from interception and unauthorized access making it suitable for secure inter-branch communication.

**Centralized authentication** was configured using Network Policy Server and Active Directory to provide authentication, authorization and accounting services on all the network infrastructure. Using this these services will ensure that user access is managed from a centralized location which will help and allow administrator to enforce consistent security policies and control access to network resources based on user roles. By centralizing authentication, the risk of unauthorized access will be reduced in addition to simplified management as all user credentials and permissions are maintained within a single directory service.

**Monitoring and Logging** mechanisms were implemented by using PRTG and Graylog to enhance and improve the network security and information monitoring. PRTG provides a real time monitoring of device status, performance and components while Graylog collects, stores and analyzes the information that been generated by the networking devices. These systems allow administrators to detect anomalies, identify potential security risks. Making it one of the most important security tools inside the network.

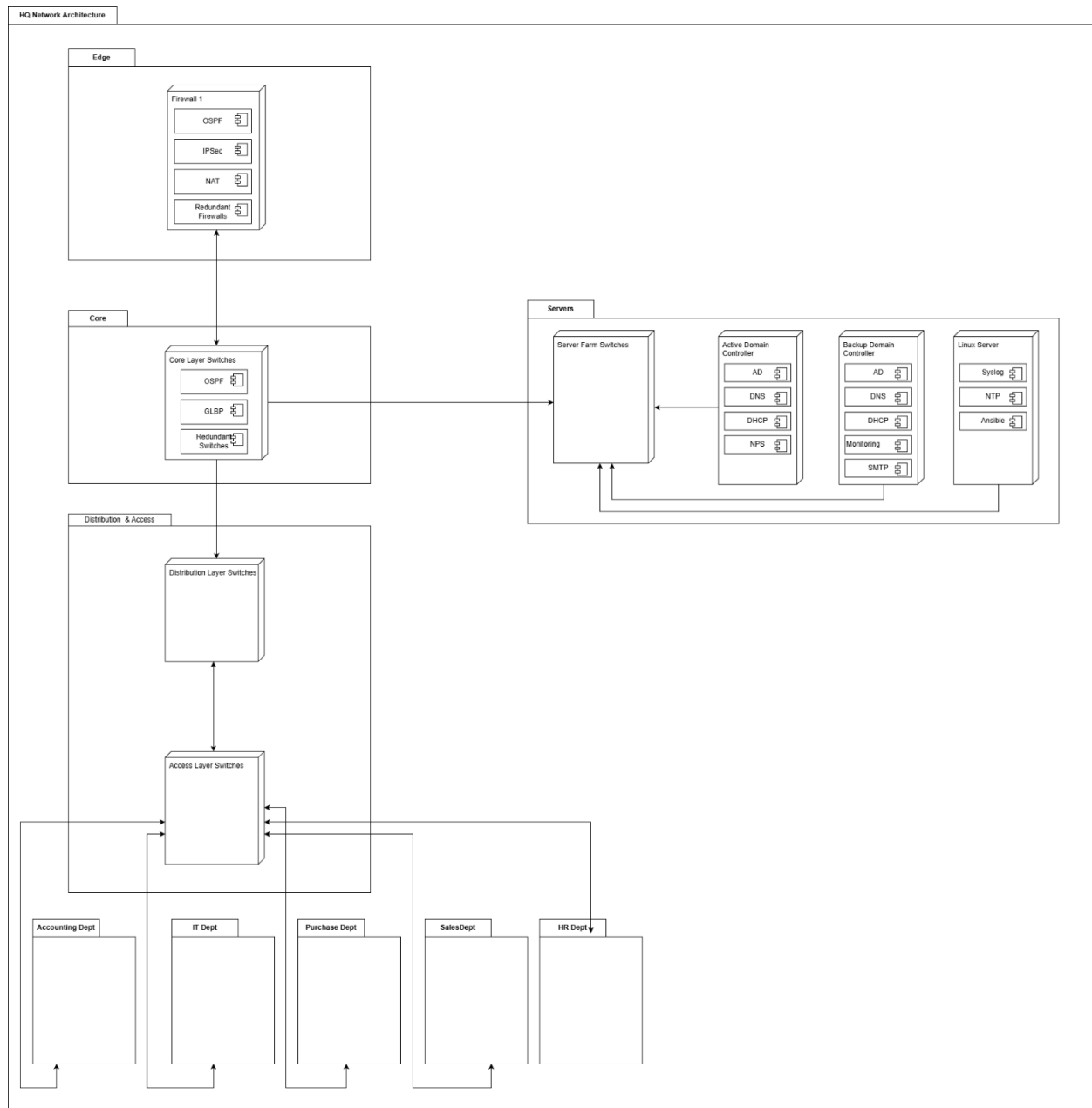
**Multi-layer Access Control Lists (ACL)** were configured to control traffic flow in multiple stages in the network to enforce security policies. These ACLs restricts and deny unauthorized communication between VLANs and external network. In addition, **VPN-Filters** were configured

and applied on the Cisco ASA to limit the type of traffic that are allowed to pass through the IPsec tunnel ensuring only authorized services and protocols can pass between sites.

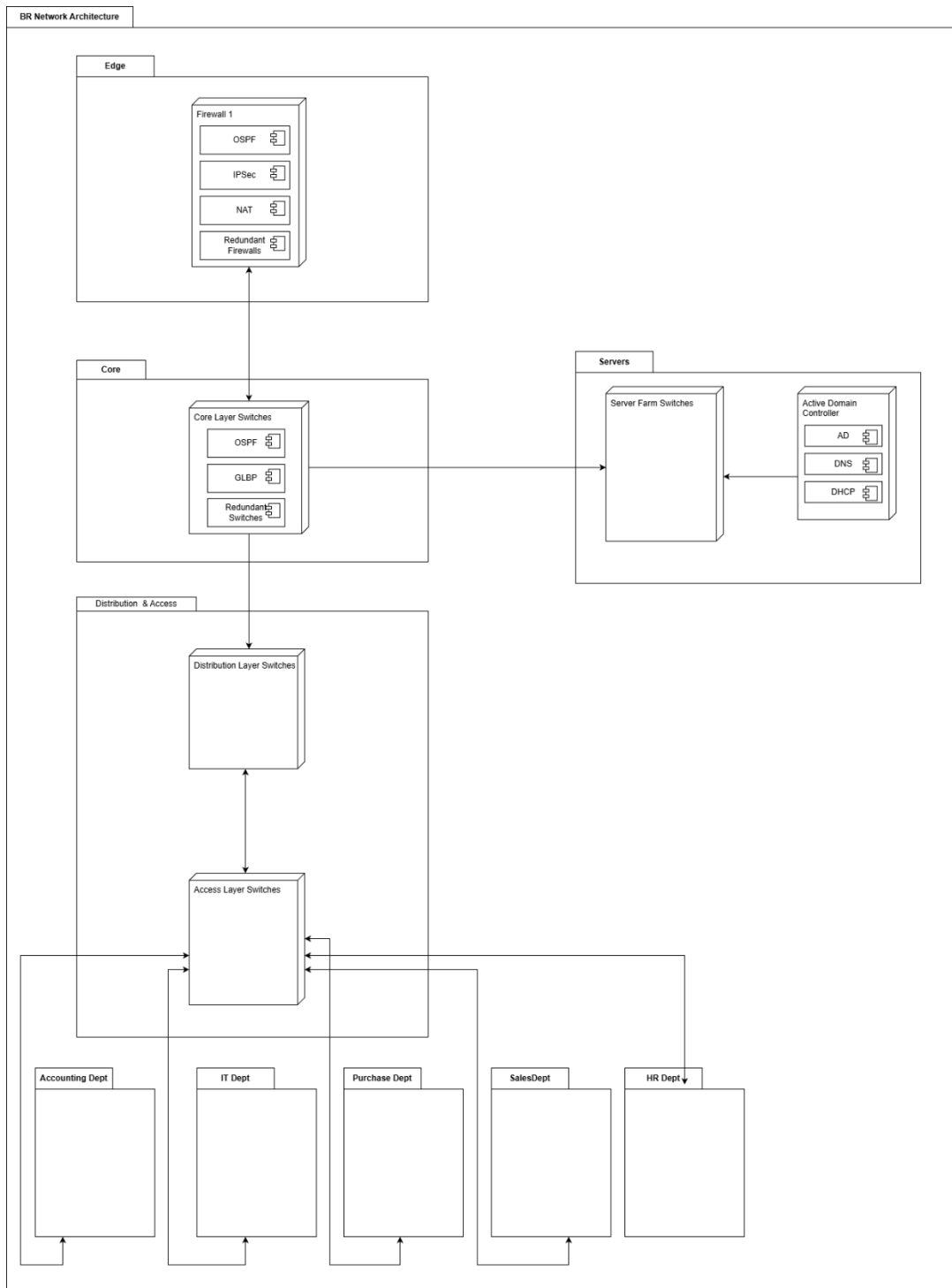
**Network and segmentation** were implemented using **VLAN** to isolate different departments and services within the company. This approach reduces the attack space by limiting the unauthorized access between the isolated segments. Segmentation enhances the security and performance by controlling the broadcast domains within the network and by enforcing logical separation between the departments and their services.

**Layer 2 Security Mechanisms** were implemented to protect the network from commonly known switching attacks. The deployed mechanisms such as BPDU Guard, Root Guard, and Loop Guard were implemented to prevent unauthorized switches from changing the spanning tree topology and to ensure network stability.

## Architecture Diagrams – HQ



## Architecture Diagrams – BR





[illegible]

## Deployment Diagram – WAN/IPSec

